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Definition

GeoPops¹ is an open-source tool for generating geographically and demographically realistic synthetic populations for any US Census area from publicly available data with associated networks for home, school, workplace, and group quarters (GQ) (dormitories, nursing homes, etc.). GeoPops has the same underlying methodology as GREASYPOP-CO² with the following changes:

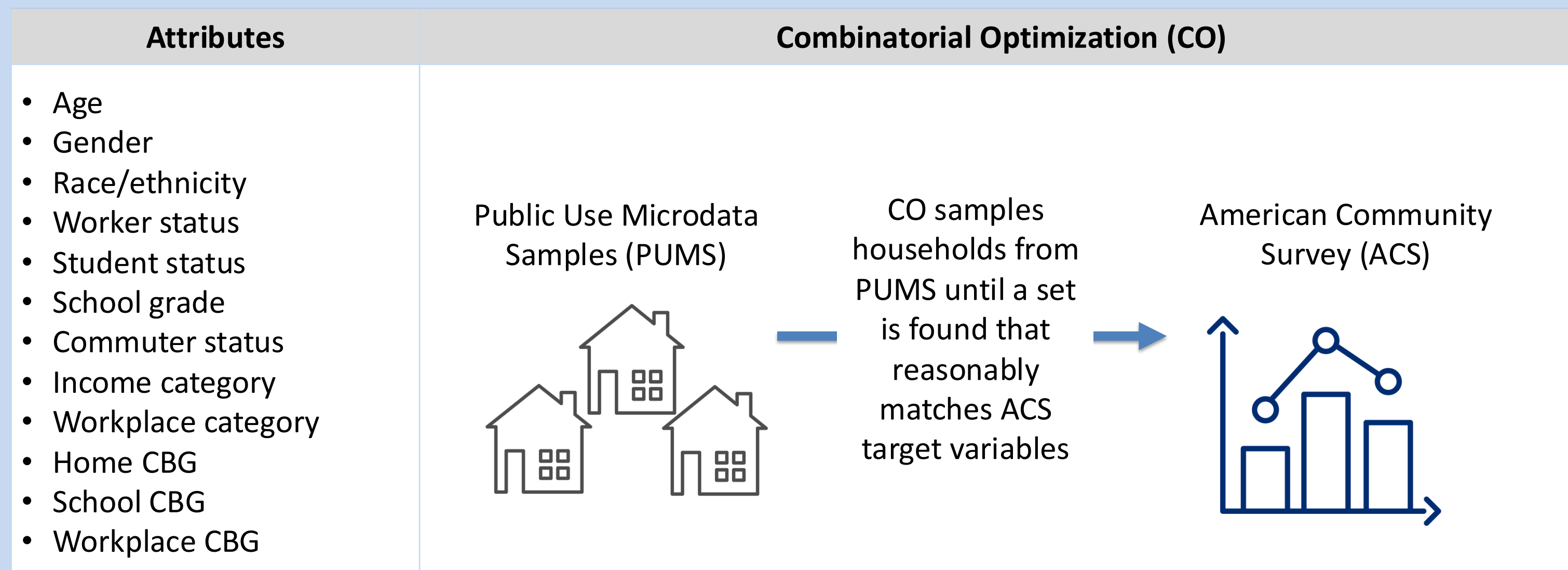
- Wrapped in convenient Python package
- Can be pip installed
- Compatibility with Census data beyond 2019 (still developing)
- Automated data downloading
- Users can adjust all config parameters from the front-end
- Compatibility with the agent-based modeling software Starsim³

github.com/ACCIDDA/GeoPops

pypi.org/project/geopops

Methodology

Step 1: Generate individuals and households



Usage

Code

Example for Howard County, MD

```
import geopops

pars = {'path': 'YOUR_OUTPUT_DIR', # Where you want outputs
        'census_api_key': 'YOUR_KEY', # Your Census API key
        'main_year': 2019, # Main year
        'geos': ['24027'], # State or county FIPS of main geo
        'commute_states': ['24'], # Commute to/from states
        'use_pums': ['24']} # PUMS states to download

geopops.WriteConfig(**pars) # Set parameters
geopops.DownloadData() # Download raw data files
geopops.ProcessData() # Preprocessing for next steps
geopops.RunJulia() # CO, network generation, export to csv
geopops.ForStarsim() # Format for Starsim
```

Step 2: School and Workplace Assignment

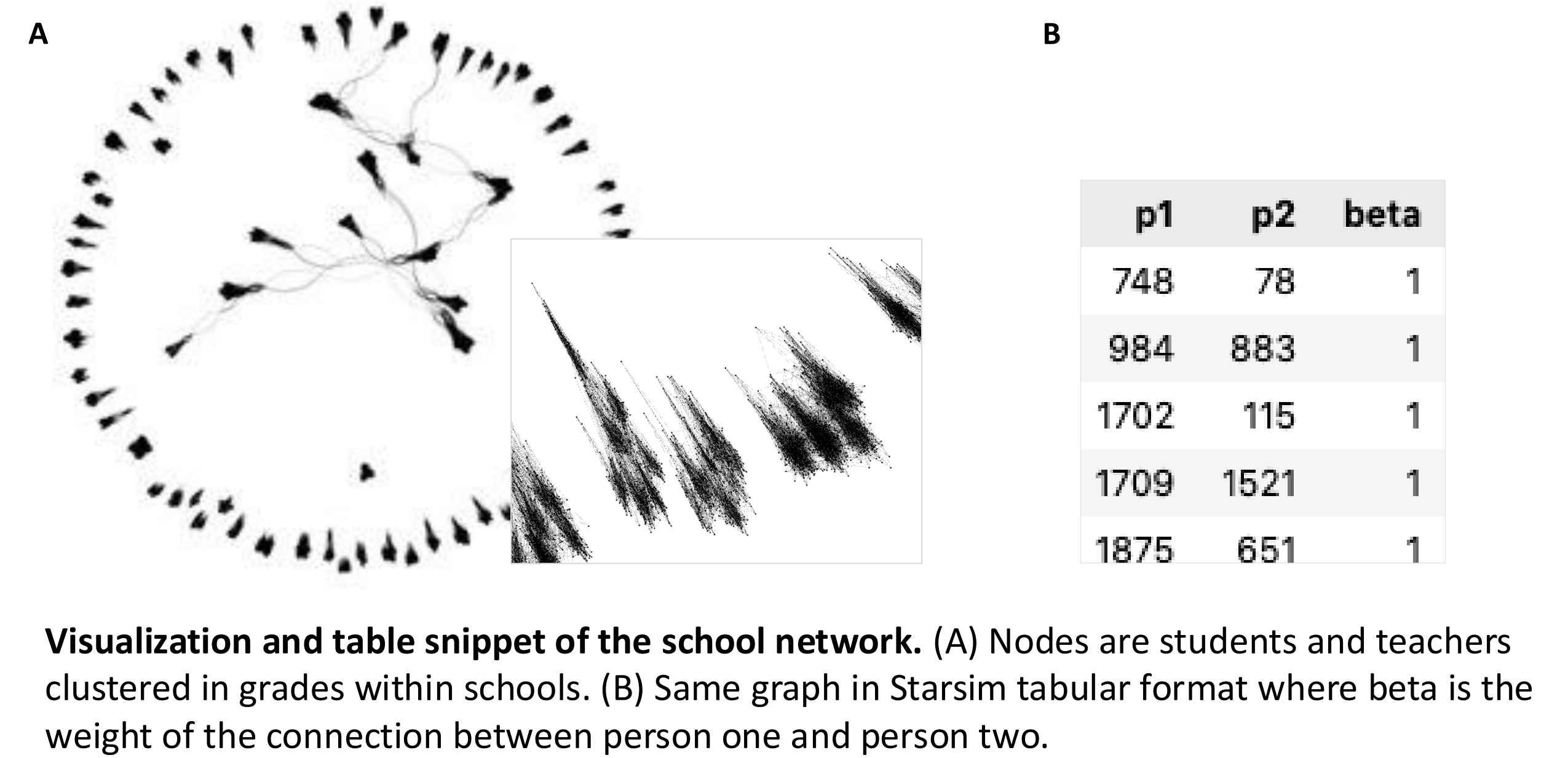
	Methods	Data	Data Sources
Schools	• Students assigned to closest school that offers required grade • Teachers assigned to school by selecting teachers from nearby CBGs	Student and staff counts for each grade level in public schools	National Center for Education Statistics (NCES)
Workplaces	• Iterative proportional fitting (IPF) to make origin-destination commute matrix • IPF to make industry by destination commute matrix for each origin CBG • Includes outside workers	Origin-destination and workplace category matrices	Longitudinal Employer-Household Dynamics (LEHD)

Outputs

- Original files from each data source
- csv of agents and attributes
- csv of homes with locations (CBG level)
- csv of schools with locations and students and teachers by grade
- csv of workplaces with locations and workers by category
- csv of GQ facilities with locations and residents and workers by type (e.g., dormitory vs LTCF)
- csv of outside workers (people who commute into area for work)
- Files in matrix and csv format for each network
- Starsim compatible files for agents and networks

Step 3: Network Generation

Network	Algorithm	Parameters
Household	All agents within a household are fully connected	• N/A
School	Stochastic Block Model within each school	• Assortativity: $\alpha=0.9$ • Mean degree: $K=12$ • Blocks are grade levels (Pre-k – 12)
Workplace	Stochastic Block Model within each workplace	• Assortativity: $\alpha=0.9$ • Mean degree: $K=8$ • Blocks are income levels ($<40k$ or $\geq 40k$)
Group quarters	Watts-Strogatz Model within each GQ facility	• Mean degree: $K=12$ • Rewiring probability: $\beta=0.25$



Example of Howard County, MD*

Components passed into Starsim to model disease transmission

GeoPops agents

GeoPops networks

SIR model

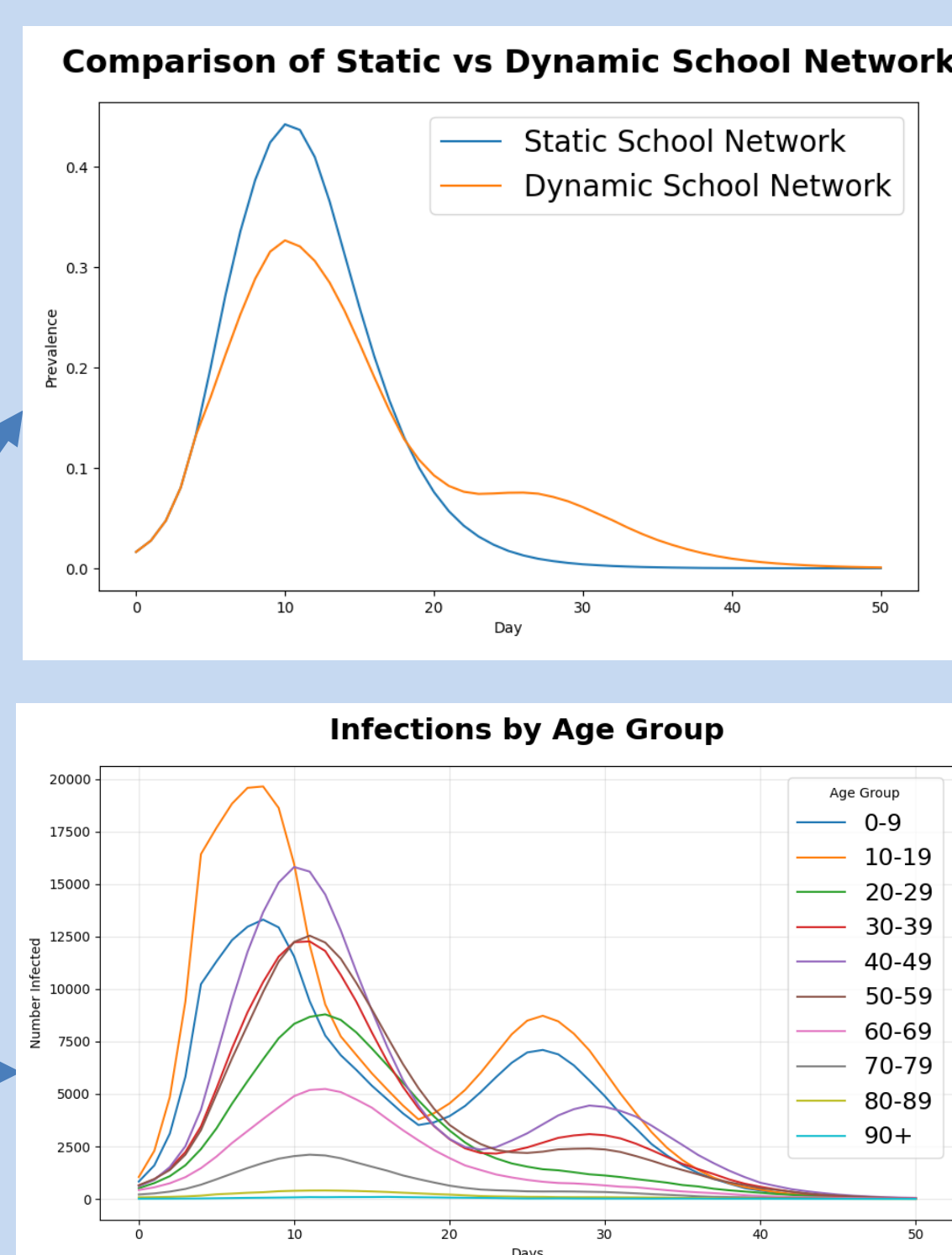
- Upper respiratory disease
- Infection duration ~7 days
- Transmission coefficient ~0.1
- 50-day simulation

Feedback module

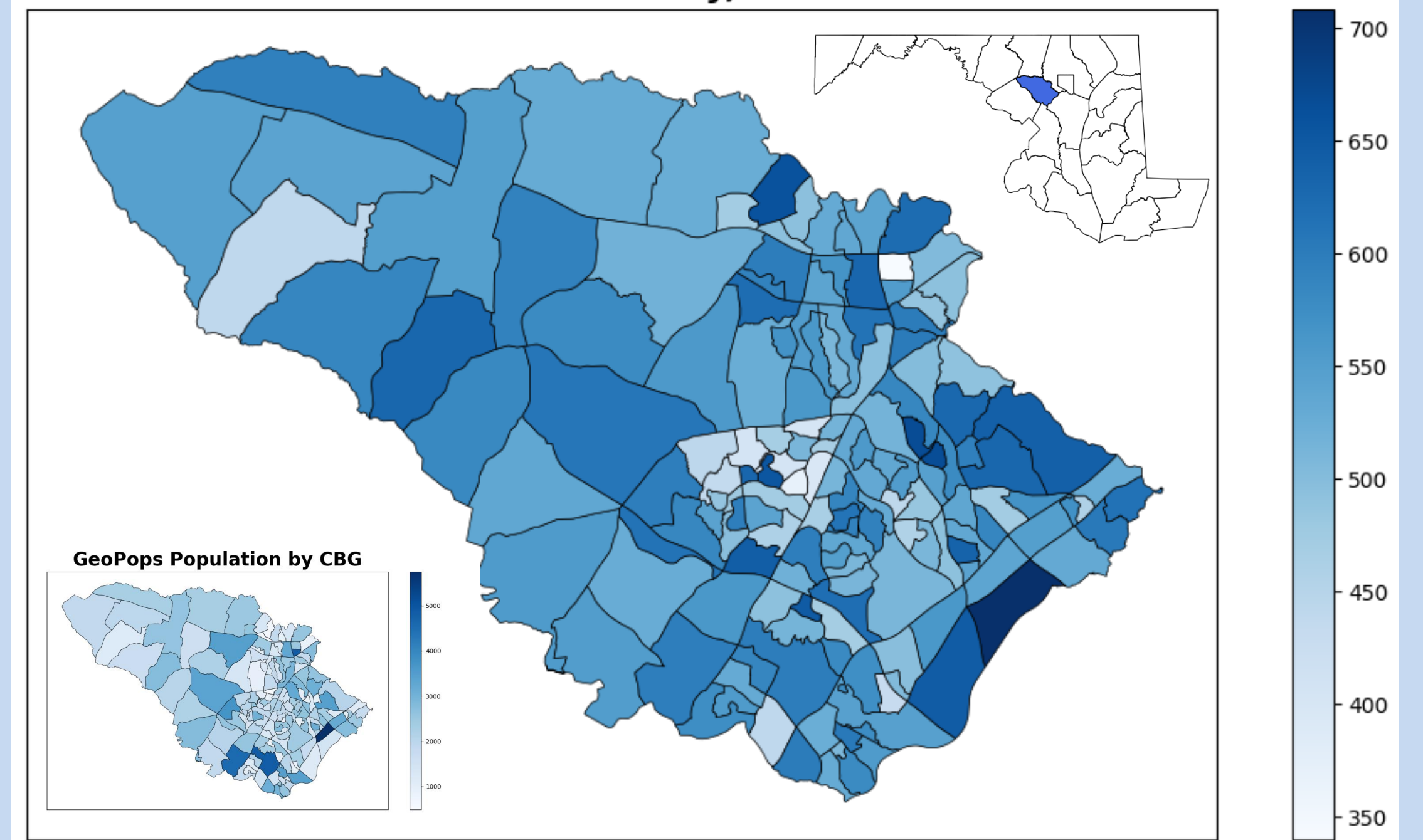
- Removes edges of school network when prevalence > 0.1

Subgroup Tracking

- Infections by age and CBG



Total Infections per 100 Population by CBG Howard County, MD



*This example reflects a simple application with a generic upper respiratory infection.

References

Hamilton, Alisa, Sasha Tulchinsky, Gary Lin, Cliff Kerr, Eili Klein, and Lauren Gardner. 2025. GeoPops: Geographically Realistic Synthetic Population Generator. Johns Hopkins University and One Health Trust. <https://github.com/ACCIDDA/GeoPops>.
Tulchinsky, Alexander, Fardad Haghighpanah, Alisa Hamilton, Nodar Kipshidze, and Eili Klein. 2024. "Generating Geographically and Economically Realistic Large-Scale Synthetic Contact Networks: A General Method Using Publicly Available Data." arXiv, ahead of print, June 20. <https://doi.org/10.48550/arXiv.2406.14698>.
Kerr, Cliff, Robyn Stuart, Romesh Abeyuriya, et al. 2024. "Starsim: A Fast, Flexible Toolbox for Agent-Based Modeling of Health and Disease." <https://starsim.org/>.

github.com/ACCIDDA/GeoPops/tutorials/MIDAS

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